Mapping of KSBs to assessment methods

KNOWLEDGE	ASSESSMENT METHODS
K1 Types of data, for example, structured, unstructured, qualitative, quantitative, numeric, strings, compound data types.	Professional discussion underpinned by a portfolio of evidence
K2 Common sources of data, for example, internal, external, open data sets, public and private.	Professional discussion underpinned by a portfolio of evidence
K3 Data storage formats and their importance for analysis, for example, relational database tables, spreadsheets, bespoke digital applications, comma separated value lists, text documents, voice and video.	Professional discussion underpinned by a portfolio of evidence
K4 Data element formats and how their selection can impact precision, analysis and communication, for example, integers, floating point numbers and their precision, scientific notation, date formatting as strings.	Scenario demonstration with questioning
K5 How to access and extract data from already identified sources.	Scenario demonstration with questioning
K6 How to collate and format data in line with organisational standards.	Scenario demonstration with questioning
K7 Why it may be important to anonymise data, for example for privacy, security and regulatory compliance, or to eliminate potential for bias.	Professional discussion underpinned by a portfolio of evidence
K8 How to anonymise data, for example one-for-one replacement of names, addresses or telephone numbers with distinct new values, without changing data structure or relationships.	Professional discussion underpinned by a portfolio of evidence

K9 Management and presentation tools to visualise and review the characteristics of data. Examples include spreadsheets with tables and charts, dashboarding tools, custom tools for particular data types, systems or contexts.	Professional discussion underpinned by a portfolio of evidence
K10 Communication tools and technologies for collaborative working, including the ability to share data and findings of data reviews. Examples include dashboards, shared whiteboards, or presentation tools for video conferencing for face-to-face contexts or digital presentation displays.	Professional discussion underpinned by a portfolio of evidence
K11 Communication methods, formats and techniques to help audiences understand data findings and their implications, for example written, verbal, non-verbal, presentation, email, conversation, storytelling and active listening.	Professional discussion underpinned by a portfolio of evidence
K12 Roles within an organisation needing access to data or to understand data findings, and how these roles impact the amount of detail needed in data communications, for example, customer, manager, peer; technical and non-technical.	Professional discussion underpinned by a portfolio of evidence
K13 How to combine data from multiple sources. For example using look ups, copy and paste and visualisation tools or data blending tools on bespoke systems.	Scenario demonstration with questioning
K14 Understand the capabilities within data analysis, visualisation, and querying tools, for example, spreadsheets or database viewers or digital display screens on bespoke systems for use in answering questions, solving problems, and the potential to use automation for repeated data manipulation.	Scenario demonstration with questioning
K15 How to filter details, focusing on information relevant to the data tasks and purpose.	Scenario demonstration with questioning

K16 Basic statistical methods to extract relevant information from structured and unstructured data, for example, counting rows, calculating the mean and standard deviation of numeric fields, counting words in a document, listing the most common values, calculating percentage contributions or percentage differences between data items.	Scenario demonstration with questioning
K17 Common data quality issues that can arise for example misclassification, duplicate entries, spelling errors, obsolete data, compliance issues and misinterpretation or translation of meaning.	Professional discussion underpinned by a portfolio of evidence
K18 Methods of validating data and the importance of taking corrective action, for example checking the source of information, identification and standardisation of outliers, adjusting item counts or totals of values.	Professional discussion underpinned by a portfolio of evidence
K19 Legal and regulatory requirements surrounding the use of data for example GDPR, Data Protection Act, data security, intellectual property rights, data sharing, marketing consent, personal data definition, and sector specific standards.	Professional discussion underpinned by a portfolio of evidence
K20 The ethical use of data, including in relation to its use with Artificial Intelligence and other automated systems, and the potential impacts of unethical use of data on the organisation.	Professional discussion underpinned by a portfolio of evidence
K21 The value of data to an organisation, for example to understand behaviours, to assess stakeholder sentiment, to interpret inputs received, to identify trends, to improve decision making and efficiency, or to build strategic or tactical plans to address a current situation.	Professional discussion underpinned by a portfolio of evidence
K22 The significance of understanding cultural awareness, diversity and accessibility with respect to data sets.	Professional discussion underpinned by a portfolio of evidence
K23	Professional discussion underpinned by a

The relationships between data, machine learning, Internet of Things (IoT), Artificial Intelligence (AI) and Generative AI. For example, the impact of data and any biases within it on training AI models, and the impact of AI on data volume, quality, security, privacy and ethical considerations.	portfolio of evidence
K24 Sustainable data practices for example organisational policies and procedures relating to environmental impact and sustainability, green data centres, and responsible data storage.	Professional discussion underpinned by a portfolio of evidence
K25 Principles and policies of equity, diversity and inclusion in the workplace and their impact on the organisation.	Professional discussion underpinned by a portfolio of evidence
K26 Understand when and how to apply the principles of prompt engineering to identify and research effective data transformation techniques to ensure data quality and integrity.	Scenario demonstration with questioning

SKILL	ASSESSMENT METHODS
S1 Select and migrate data from already identified sources.	Scenario demonstration with questioning
S2 Format and save datasets.	Scenario demonstration with questioning
S3 Summarise, analyse and explain gathered data.	Scenario demonstration with questioning
S4 Combine data sets from multiple sources and present in format appropriate to the task.	Scenario demonstration with questioning
S5 Use tools and/or apply basic statistical methods to identify trends and patterns in data.	Scenario demonstration with questioning
S6 Identify faults and cleanse data to improve data quality, for example identifying gaps, duplicate entries, outliers and unusual variances, including cross-checking across data elements or between data sources.	Scenario demonstration with questioning
S7 Audit data results for maintenance of data quality, reviewing a data set once all sources are combined, to ensure accuracy, completeness, consistency and traceability from original data.	Scenario demonstration with questioning
S8 Demonstrate the different ways of communicating meaning from data in line with audience requirements.	Professional discussion underpinned by a portfolio of evidence
S9 Produce clear and consistent documentation of the data provided to others and of actions completed. Where appropriate or mandated by the working context, this documentation should use standard organisational templates.	Professional discussion underpinned by a portfolio of evidence
S10	Professional discussion underpinned by a

Store, manage and distribute data in compliance with organisational, national, sector specific standards and or legislation.	portfolio of evidence
S11 Considers sustainability and ways to reduce impact. For example, using cloud storage, sharing links to files, avoid storing multiple versions of files, and reducing the use of physical handouts of documentation.	Professional discussion underpinned by a portfolio of evidence
S12 Parse data against standard formats, and test and assess confidence in the data and its integrity.	Professional discussion underpinned by a portfolio of evidence
S13 Operate collaboratively in a working context that accounts for, and takes advantage of, the roles, skills and activities of others, especially those interacting with the same data sets or working towards a common goal.	Professional discussion underpinned by a portfolio of evidence
S14 Prioritise own activities within the context of the duties to be performed, taking account of any known or expected impact on others.	Professional discussion underpinned by a portfolio of evidence
S15 Follows equity, diversity and inclusion policies in the organisation for a common goal.	Professional discussion underpinned by a portfolio of evidence
S16 Demonstrate the ability to use different tools and methods to formulate and utilise effective prompts to research, apply, and evaluate data transformation techniques.	Scenario demonstration with questioning

BEHAVIOUR	ASSESSMENT METHODS
B1 Manage own time to meet deadlines and manage stakeholder expectations whether working independently or in a multidisciplinary team.	Professional discussion underpinned by a portfolio of evidence
B2 Work independently and methodically.	Professional discussion underpinned by a portfolio of evidence
B3 Support social inclusion in the workplace. For example consider the needs of the audience.	Professional discussion underpinned by a portfolio of evidence
B4 Takes responsibility for acting sustainably in their role for example switching off lights and systems when not in use, reducing file size and attachments on emails, and recycling.	Professional discussion underpinned by a portfolio of evidence

Mapping of KSBs to grade themes

Scenario demonstration with questioning

KSBS GROUPED BY THEME	KNOWLEDGE	SKILLS	BEHAVIOUR
Data gathering K4 K5 K6 K13 S1 S2 S3 S4	Data element formats and how their selection can impact precision, analysis and communication, for example, integers, floating point numbers and their precision, scientific notation, date formatting as strings. (K4) How to access and extract data from already identified sources. (K5) How to collate and format data in line	Select and migrate data from already identified sources. (S1) Format and save datasets. (S2) Summarise, analyse and explain gathered data. (S3) Combine data sets from multiple sources and present in format appropriate to the task. (S4)	None

	with organisational standards. (K6) How to combine data from multiple sources. For example using look ups, copy and paste and visualisation tools or data blending tools on bespoke systems. (K13)		
Data analysis and validation K14 K15 K16 K26 S5 S6 S7 S16	Understand the capabilities within data analysis, visualisation, and querying tools, for example, spreadsheets or database viewers or digital display screens on bespoke systems for use in	Use tools and/or apply basic statistical methods to identify trends and patterns in data. (S5) Identify faults and cleanse data to improve data quality, for example identifying gaps,	None

	answering questions, solving problems, and the potential to use automation for repeated data manipulation. (K14) How to filter details, focusing on information relevant to the data tasks and purpose. (K15) Basic statistical methods to extract relevant information from structured and unstructured data, for example, counting rows, calculating the mean and standard deviation of numeric fields, counting words in a document, listing the most common values, calculating percentage contributions or percentage differences between data items. (K16) Understand when and how to apply the principles of prompt engineering to identify and research effective data transformation techniques to ensure data quality and integrity. (K26)	duplicate entries, outliers and unusual variances, including cross-checking across data elements or between data sources. (S6) Audit data results for maintenance of data quality, reviewing a data set once all sources are combined, to ensure accuracy, completeness, consistency and traceability from original data. (S7) Demonstrate the ability to use different tools and methods to formulate and utilise effective prompts to research, apply, and evaluate data transformation techniques. (S16)		

Professional discussion underpinned by a portfolio of evidence

KSBS GROUPED BY THEME	KNOWLEDGE	SKILLS	BEHAVIOUR
Data analysis and validation K17 K18 S12	Common data quality issues that can arise for example misclassification, duplicate entries, spelling errors, obsolete data, compliance issues and misinterpretation or translation of meaning. (K17) Methods of validating data and the importance of taking corrective action, for example checking the source of information, identification and standardisation of outliers, adjusting item counts or totals of values. (K18)	Parse data against standard formats, and test and assess confidence in the data and its integrity. (S12)	None
Distribution and dissemination K9 K10 K11 K12 K19 K21 K22 S8 S9 S10	Management and presentation tools to visualise and review the characteristics of data. Examples include spreadsheets with tables and charts, dashboarding tools, custom tools for particular data types, systems or contexts. (K9) Communication	Demonstrate the different ways of communicating meaning from data in line with audience requirements. (S8) Produce clear and consistent documentation of the data provided to others and of actions completed. Where appropriate or mandated by the	None

	tools and technologies for	working context, this documentation	

	collaborative working, including the ability to share data and findings of data reviews. Examples include dashboards, shared whiteboards, or presentation tools for video conferencing for face-to-face contexts or digital presentation displays. (K10) Communication methods, formats and techniques to help audiences understand data findings and their implications, for example written, verbal, non-verbal, presentation, email, conversation, storytelling and active listening. (K11) Roles within an organisation needing access to data or to understand data findings, and how these roles impact the amount of detail needed in data communications, for example, customer, manager, peer; technical and non-technical. (K12) Legal and regulatory	should use standard organisational templates. (S9) Store, manage and distribute data in compliance with organisational, national, sector specific standards and or legislation. (S10)		

	requirements surrounding the use of data for example GDPR, Data Protection Act, data security, intellectual property rights, data sharing, marketing consent, personal data definition, and sector specific standards. (K19) The value of data to an organisation, for example to understand behaviours, to assess stakeholder sentiment, to interpret inputs received, to identify trends, to improve decision making and efficiency, or to build strategic or tactical plans to address a current situation. (K21) The significance of understanding cultural awareness, diversity and accessibility with respect to data sets. (K22)		
Approaches to work K24 K25 S11 S13 S14 S15 B1 B2 B3 B4	Sustainable data practices for example organisational policies and procedures relating to environmental impact and sustainability, green	Considers sustainability and ways to reduce impact. For example, using cloud storage, sharing links to files, avoid storing multiple versions of	Manage own time to meet deadlines and manage stakeholder expectations whether working independently or in a multidisciplinary team. (B1)

	data centres, and responsible data storage. (K24) Principles and policies of equity, diversity and inclusion in the workplace and their impact on the organisation. (K25)	files, and reducing the use of physical handouts of documentation. (S11) Operate collaboratively in a working context that accounts for, and takes advantage of, the roles, skills and activities of others, especially those interacting with the same data sets or working towards a common goal. (S13) Prioritise own activities within the context of the duties to be performed, taking account of any known or expected impact on others. (S14) Follows equity, diversity and inclusion policies in the organisation for a common goal. (S15)	Work independently and methodically. (B2) Support social inclusion in the workplace. For example consider the needs of the audience. (B3) Takes responsibility for acting sustainably in their role for example switching off lights and systems when not in use, reducing file size and attachments on emails, and recycling. (B4)
Data Gathering K1 K2 K3 K7 K8 K20 K23	Types of data, for example, structured, unstructured, qualitative, quantitative, numeric, strings, compound data types. (K1) Common sources of data, for	None	None

	example, internal, external, open data sets, public and private. (K2) Data storage formats and their importance for analysis, for example, relational database tables, spreadsheets, bespoke digital applications, comma separated value lists, text documents, voice and video. (K3) Why it may be important to anonymise data, for example for privacy, security and regulatory compliance, or to eliminate potential for bias. (K7) How to anonymise data, for example one-for-one replacement of names, addresses or telephone numbers with distinct new values, without changing data structure or relationships. (K8) The ethical use of data, including in relation to its use with Artificial Intelligence and other automated systems, and the potential impacts of unethical use of			

	data on the organisation. (K20) The relationships between data, machine learning, Internet of Things (IoT), Artificial Intelligence (AI) and Generative AI. For example, the impact of data and any biases within it on training AI models, and the impact of AI on data volume, quality, security, privacy and ethical considerations. (K23)		

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